A protocol for qualifying robot telemetry before interpreting motion metrics

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Abstract

A numerical summary of robot telemetry can be valid for a recorded interval while being inadequate for the physical claim attached to it. We describe an offline protocol that specifies the quantity and required events, checks signal meaning and observation support, and reports which conclusions remain unavailable. The protocol includes exact-count sampling comparisons that distinguish observation budget from temporal placement. In 15 archived waist records, post-entry sampling retained a median 0.331 of the enable-inclusive reported-velocity peak. At the same sample budget, first-observation, endpoint and uniform-index allocations retained medians of 1.000, 0.363 and 0.734. The first-observation allocation was therefore a strong baseline in this startup series. A separate application used all 50 episodes and 55,000 numeric rows of a public ALOHA dataset. With five retained observations, endpoint and uniform-index allocations had identical time extents but retained median state-excursion fractions of 0.389 and 0.903. These results support comparing allocation rules for a specified quantity rather than selecting a universal window. Public data remain ineligible for activation-response, internal-reference tracking and source-freshness claims. The procedure and public-only implementation are supplied for reuse.

Value of the method

- The procedure separates a computable recorded-data quantity from the additional evidence required for an activation, tracking, completion or freshness claim.
- Exact-count comparisons expose effects of sample placement, including comparisons with both the same sample count and the same first and last observation times.
- A schema-independent implementation and a complete public numeric example permit reproduction without access to the private robot development archive.

Specifications table

Item	Specification
Subject area	Engineering
More specific subject area	Robot telemetry and measurement procedures
Name of the method	Qualification of robot execution telemetry for a specified transient-assessment query
Original method	A workflow combining established event-based summaries, provenance checks and deterministic subsampling. No new control law or measurement standard is claimed.
Resource availability	Offline Python implementation and complete public numeric example; pinned ALOHA revision and source hashes. Restricted historical records require author-approved access.

Background

Robot execution assessment often begins with an available signal rather than an explicitly defined quantity. A peak velocity, command–state difference or advancing timestamp may then acquire a stronger interpretation than its recording supports. Metrology distinguishes the quantity intended to be measured from the procedure and observations used to estimate it [1]. Reusable data and transparent procedures are also central to FAIR data management [2] and reproducible robotics [3]. These principles motivate an operational check before an engineering log is used to support a transient-response claim.

Reference generation and fault diagnosis answer different questions. A jerk-limited trajectory generator constrains its reference [4]; it does not establish that an archived record covers every enable transition. Fault diagnosis requires additional models, assumptions or labels [5]. Likewise, a stored client command need not be a synchronised internal drive reference. The present method does not replace these methods or introduce a new control law. It adapts general measurement and reproducibility principles into a sequence of query-specific evidence checks for robot telemetry.

Two practical distinctions motivate the validation. First, a comparison using a longer observation interval may confound temporal placement with sample budget. Second, a rule that captures an activation peak in one archive may be unsuitable for a task-wide position excursion. We therefore compare deterministic allocations with exactly matched observation counts and retain unfavourable comparisons. A public bimanual dataset provides a separate application of the same allocation code, using a quantity appropriate to its available channels. This application tests computational portability, not replication of the waist's activation response.

Method details

1 Define the question before choosing the window

Write a query containing the signal, units, interval, required events and intended interpretation. Distinguish a recorded peak from a continuous-time bound, an observed terminal phase from physical task success, and a source timestamp from a software read label. Table 1 gives the requirements used here. A missing requirement produces an unavailable or unknown conclusion; it does not imply that the robot failed.

Table 1. Query-specific evidence requirements. Supported refers to the stated recorded-data query; no row establishes physical safety or a continuous-time bound.

Query	Required evidence	Meaning when missing
Recorded extremum	Named channels, finite values, units or per-channel normalisation, and observed support	The corresponding quantity is unavailable; never fill an absent signal with zero.
Activation interval	Ordered enable/entry markers, bracketing observations and a declared gap limit	No complete activation-interval statement; a post-entry peak can remain computable.
Internal tracking	Documented internal reference and synchronised feedback	A client command–state difference does not supply this quantity.
Action completion	A native terminal event with its documented scope	Recording end alone does not establish physical task success.
Source freshness	Independent source evidence and documented clock relationship	Monotone software labels are insufficient.

For the local activation query, tE is the earliest logged enable_transition and tF is the earliest explicit first_feedback. The requested interval extends from tE to tF + 100 ms. These are software observation markers, not independently measured brake-release or sensor-acquisition instants. Other wrappers may

expose different events. An adapter must preserve those meanings instead of renaming a convenient event as first_feedback.

The public query concerns each channel's recorded state excursion from its first observed value over the whole episode. Its 14-dimensional state includes revolute-joint and gripper channels [6,7]. We therefore compute a retained fraction within each channel before aggregation. We do not take a maximum across their raw magnitudes or derive an undocumented joint-velocity signal.

2 Map the source to a canonical evidence record

Retain the original bytes and a manifest containing the source, revision where available, byte count and SHA-256. Map signal columns and events explicitly. Record the origin and units of each clock, whether a command is a client value or internal reference, and what a terminal flag means. Leave undocumented properties null. The JSON reader rejects string-valued booleans, non-integer nanosecond fields and reversed observation bounds. The assessment imports no robot SDK and requires no connection to a robot.

The supplied reference implementation represents activation bounds, observed support, maximum gap, finite-state availability, reference provenance, terminal-event evidence and source-clock evidence separately. The core function evaluates each query independently. A Boolean provenance field is an audited input assertion; setting it to true is not itself verification. Its supporting source or acquisition documentation must be retained alongside the manifest. Figure 1 summarises the procedure.

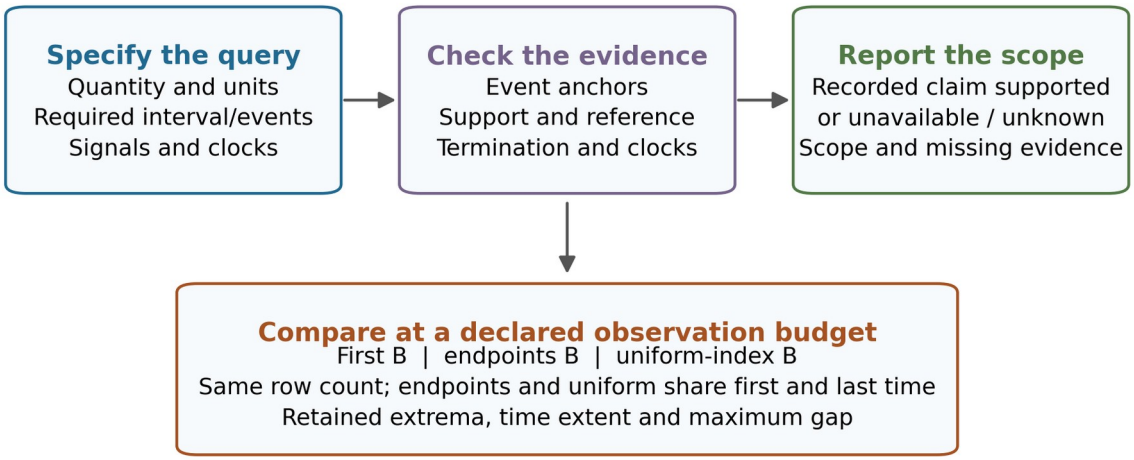


Figure 1. Offline procedure for a specified robot-telemetry query. Required events, signal roles and source-clock evidence are checked separately. Exact-count allocations compare extrema only after the quantity and recorded support have been defined. An unavailable claim is not a physical failure label.

3 Check observation support and clock meaning

Sort observations by the declared analysis time and retain original row order for ties. Audit non-finite values, duplicated labels, conflicting states under a shared label, observed time extent and inter-observation gaps. Do not interpolate to reconstruct an unobserved maximum. Keep duplicate-state counts separate from duplicate-time counts: a stationary or quantised state can repeat without stale acquisition, and separately stamped reads can reuse an underlying image.

An activation query requires ordered anchors, finite required channels and observations bracketing the requested interval. We apply a declared maximum-gap limit to the original source interval, not the subsampled allocations. The local illustration uses 20 ms, with 5 and 10 ms sensitivity. These reporting thresholds are neither validated safety limits nor guarantees of physical bandwidth. Passing the source check supports a recorded-extrema query; a selected subset does not inherit this status.

For internal tracking, require a documented internal reference and its synchronisation with feedback. For action completion, identify the native terminal event and state its scope. For freshness, require independent source evidence and a documented clock relationship. Monotone software timestamps, a fixed nominal frequency or an episode-ending flag do not meet all of these requirements by themselves. Unknown properties must remain visible in the report.

4 Compare allocations at a declared sample budget

Let W contain the eligible observations for a query. For local reported velocity, define P(W) as the largest absolute value over its four waist channels. For public state channel j, define $D_j(W)$ as the largest absolute change from that episode's first observed state $q_{0,j}$. The anchor $q_{0,j}$ is fixed before subsampling. The retained quantities are

$$R_v = \frac{P(W_B)}{P(W)} \quad (1a)$$

$$R_{q,j} = \frac{D_j(W_B)}{D_j(W)} \quad (1b)$$

A ratio is unavailable when its full-record denominator is zero. The reference uses all observations in W and is an empirical reference, not physical ground truth. A subset maximum cannot exceed this reference by construction. The empirical comparisons concern the different retained fractions of non-nested, equally sized subsets.

The local reference budget B is the number of observations in [tF, tF + 100 ms]. We compare the original post-entry subset with three allocations of exactly B distinct rows from W. First B takes the first B rows. Endpoints takes the first ceiling(B/2) and last floor(B/2) rows. Uniform index takes evenly spaced row indices, including the first and last. All rules operate without inspecting velocity or state magnitudes. Endpoints and uniform index therefore have the same count and temporal extent, while their interior coverage differs. Uniform index is not necessarily uniform in physical time when observation density varies.

The public example uses B = 5, 10, 25, 50 and 100 observations per episode. Five observations are used as the smallest reported budget, not as an equivalent measurement duration across tasks. At the declared 50 Hz, a contiguous five-row subset spans about 80 ms, whereas a distributed five-row subset can span the full episode. Endpoints and uniform index both retain the full time extent. Every comparison reports the actual span and maximum gap; equal count alone does not make two observation supports equivalent.

5 Report a qualified result rather than a global quality score

For each query, return the value if computable, the supporting interval and sample count, missing requirements and the permitted interpretation. Preserve numerical summaries that are meaningful at file level even when a stronger activation or tracking query is unavailable. Avoid a single pass/fail label for the robot. A dataset may support an episode-level state excursion and simultaneously lack the information needed for activation response, internal tracking or physical success.

The reusable files are telemetry_qualification.py, qualification_io.py and public_demo.py. The public-only example reads the pinned numeric data and writes per-episode allocation results. The complete reproduction scripts also rebuild the local comparisons and the independent checks. Supplementary material gives the schema mapping, allocation definitions, per-record results and example manifests.

Method validation

Archived waist data and exact-count comparison

The primary archive contains 31 eligible high-level physical records with 232,735 rows from one robot on 5 September 2026. The activation comparison uses all 15 hold records with both required markers. The remaining records, simulation self-test, recentring files and nested count-domain views remain inventoried; absent markers are not imputed. The paired records cover ten runtime-library digests. Digests identify binary contents and are neither source commits nor independent platforms. Conditions reflect sequential development, not randomised controller comparisons.

At the reference budget, the median number retained was 149. Post-entry sampling retained a median 0.331 of the enable-inclusive peak. First B retained 1.000, endpoints 0.363 and uniform index 0.734 (Table 2; Figure 2). Uniform index exceeded endpoints in ten records, was lower in two and equal in three. Their median paired retained-fraction difference was 0.253. Those allocations share count and first/last times, while their interior placement differs within the recorded support.

Table 2. All 15 local records at the reference observation budget. B equals each record's post-entry observation count (median 149). Every allocation uses exactly B rows. Endpoints and uniform index share the same first and last times. Retained fractions refer to the full observed enable-inclusive peak.

Allocation	Median retained	Minimum retained	Records ≥95%	Median extent (s)
Post-entry	0.331	0.147	3	0.100
First B	1.000	0.653	14	0.304
Endpoints	0.363	0.188	4	1.233
Uniform index	0.734	0.187	6	1.233

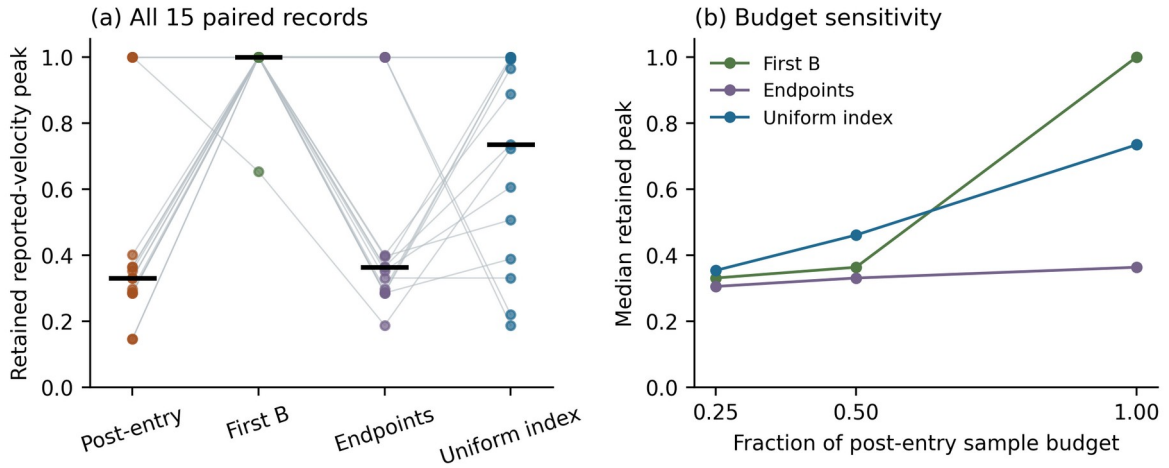


Figure 2. Exact-count local comparisons. (a) All 15 records at the reference B, linked across allocations; black bars show record medians. Endpoints and uniform index have equal count and temporal extent. First B is a strong alternative in this startup series. (b) Full record-median sensitivity at one-quarter, one-half and full budget. Lines link deterministic settings and are not confidence intervals.

First B was a strong baseline: it retained at least 95% of the reference peak in 14 records, compared with six for uniform index. Its median time extent was 0.304 s, whereas the post-entry subset spanned approximately 0.100 s. This result limits the interpretation of a budget-only comparison. In this startup series, allocating samples early often captured the intervening transient. It does not show that an early prefix will suffice for later task motion or an unobserved transient. Lower-budget results and every individual record are supplied in the supplement.

The declared 20 ms gap limit admitted all 15 original activation intervals, while the 5 and 10 ms limits each admitted eleven. Thus, the evidence-support threshold changes which intervals are qualified, even though their sampled peaks remain computable. A qualified interval still lacks independent physical freshness verification. The supplied negative controls remove a required marker, censor the pre-entry interval or assert a gap beyond the declared limit. They verify the rule implementation rather than measuring physical fault-detection accuracy.

Separate application to public bimanual data

We used the complete numeric release of lerobot/aloha_static_coffee at revision b144896feb1f37398a862927b22cd3abdf005a6b [7]. The release contains 50 episodes, 55,000 numeric rows and 14 state channels. All numeric data and metadata files were retrieved and hashed; video files were excluded. Episode IDs, lengths and frame ranges agree with the metadata, and all 50 state sequences have distinct byte hashes. The supplement documents the format-version discrepancy in the README.

The public application uses the same frozen allocation function without fitting a threshold or changing a rule to improve external results. Only the schema mapping and declared quantity change. For each episode, we first compute the retained state-excursion fraction in each channel and then its median across 14 channels. We summarise those 50 episode medians. Channels are dependent observations within an episode and are not treated as independent trials. None of the channel denominators was zero.

At B = 5, the median of episode channel medians was 0.003 for first B, 0.389 for endpoints and 0.903 for uniform index (Figure 3). Uniform index exceeded endpoints in all 50 episode summaries, with a median paired difference of 0.503. The same-count, same-extent comparison was therefore computable under this separate schema and movement regime. At B = 10, the corresponding uniform-index summary was 0.983. Full budget curves and channel-level values are retained, including weakly retained channels; a high episode median does not imply adequate observation of every channel.

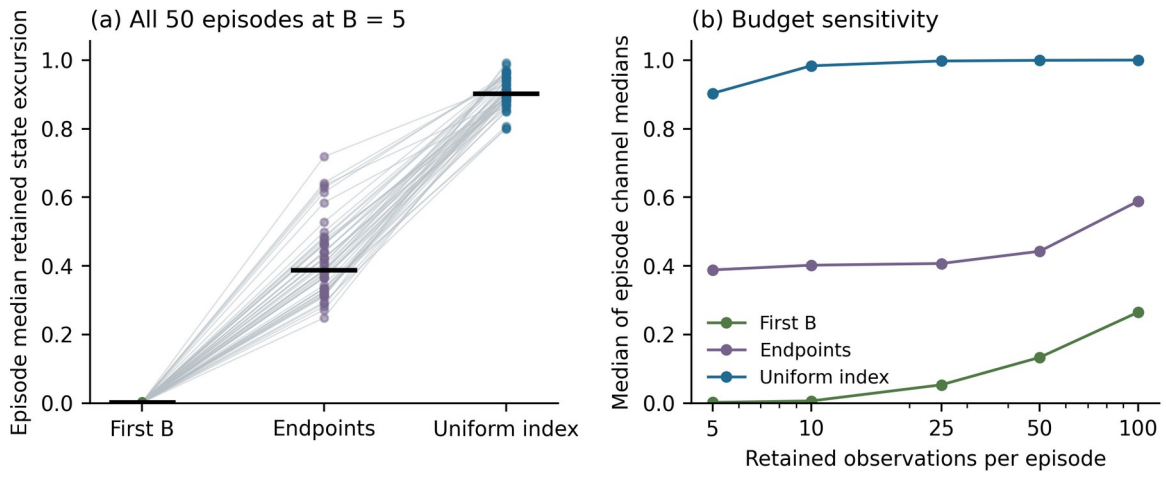


Figure 3. Public application to all 50 ALOHA episodes. (a) At B = 5, every point is a within-episode median over its 14 channel fractions; lines connect the same episode. Black bars show medians across episodes. Endpoints and uniform index share both count and time extent. (b) Complete budget sensitivity. The measured quantity is state excursion within each channel, not waist velocity, physical task success or internal tracking error.

The public release has no velocity channel or primary activation markers. Its timestamps numerically agree with frame_index/50 within 0.92 microseconds, which does not establish their hardware acquisition origin. Each episode ends with a recorded next.done flag, but no independent source establishes a physical task outcome or stop event. Activation response, internal-reference tracking and source freshness therefore remain unavailable. This is computational portability of the assessment procedure, not a controlled cross-platform activation experiment.

Computational checks and interpretation

An independent implementation uses standard-library scalar arithmetic and rational index construction to recompute local allocations and all 10,500 public channel–budget–rule results. It agrees with the primary implementation within the declared floating-point tolerance. The software checks also cover source hashes, episode counts, query gates and support thresholds. They do not add physical observations or supply an external motion reference.

The local and public results favour different allocations. The procedure makes this dependence explicit by linking each summary to its quantity, observation support and unavailable interpretations. It provides a practical addition to an unqualified peak, timestamp or command-difference summary.

Limitations

The local activation response remains a single-platform, single-day development series without a calibrated external motion reference. Software variants, guards and acquisition paths changed during development. Missing historical payload, calibration and source-to-binary mapping prevent causal comparisons and fleet reliability estimates. The public application adds an independently hosted bimanual dataset but does not supply matching enable events, internal references or controlled activation conditions.

Exact-count comparisons control the number of retained rows, and the endpoint/uniform comparison also controls temporal extent. They do not control independent sensor-image count, effective bandwidth or the physical information carried by each row. Uniform index depends on recorded row density. All retained fractions refer to finite stored observations; none bounds a continuous-time extremum. Fractions, threshold choices and digest omissions are descriptive sensitivity calculations, not population confidence intervals.

The provenance assertions supplied to the qualifier require documentary support and human review. Software cannot recover an undocumented command meaning or authenticate historical hardware from a current hostname. Conversely, an unavailable query may be a conservative consequence of missing documentation rather than a physically defective record. The method is useful for aligning claims with evidence; it does not establish fault causation, safety acceptance, task success or controller superiority.

Ethics statements

The present work reanalysed existing robot records and publicly released numeric data. No new robot operation, human-participant experiment or animal experiment was conducted. [Authors to confirm applicable institutional requirements and rights to analyse and share the historical local records.]

CRediT authorship contribution statement

[Authors to assign and confirm conceptualisation, historical data acquisition, methodology, software, validation, analysis, visualisation, writing and final approval.]

Declaration of competing interest

[Authors to confirm employment, commercial relationships and any other relevant interests.]

Funding

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Data availability

The public numeric input is available at the pinned dataset revision [7]. The accompanying public-only reproduction package contains its source manifest, numeric files, analysis code and derived tables. Historical local records and their manifests are supplied in the author review package, subject to

confirmation of sharing permissions. [Authors to provide a persistent code/data repository identifier, licence and the permitted access mechanism for restricted records before submission.]

Related research article

None. This method article is prepared as a standalone submission. [Authors to confirm related manuscripts, prior dissemination and data overlap before submission.]

References

[1] Joint Committee for Guides in Metrology. International vocabulary of metrology — Basic and general concepts and associated terms (VIM), 3rd edition, JCGM 200:2012. BIPM (2012). https://www.bipm.org/documents/20126/2071204/JCGM_200_2012.pdf (accessed 20 October 2026).

[2] Mark D. Wilkinson, Michel Dumontier, IJsbrand Jan Aalbersberg, Gabrielle Appleton, Myles Axton, Arie Baak, et al. The FAIR Guiding Principles for scientific data management and stewardship. Scientific Data 3 (2016) 160018. https://doi.org/10.1038/sdata.2016.18

[3] F. Bonsignorio. A New Kind of Article for Reproducible Research in Intelligent Robotics [From the Field]. IEEE Robot. Autom. Mag. 24 (2017) 178-182. https://doi.org/10.1109/MRA.2017.2722918

[4] L. Berscheid, T. Kröger. Jerk-limited Real-time Trajectory Generation with Arbitrary Target States. Robotics: Science and Systems XVII (2021). https://doi.org/10.15607/RSS.2021.XVII.015

[5] E. Khalastchi, M. Kalech. Fault Detection and Diagnosis in Multi-Robot Systems: A Survey. Sensors 19 (2019) 4019. https://doi.org/10.3390/s19184019

[6] Tony Z. Zhao, Vikash Kumar, Sergey Levine, Chelsea Finn. Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware. Robotics: Science and Systems XIX (2023). https://doi.org/10.15607/RSS.2023.XIX.016

[7] LeRobot contributors. ALOHA static coffee dataset, revision b144896feb1f37398a862927b22cd3abdf005a6b. Hugging Face dataset repository (2025). https://huggingface.co/datasets/lerobot/aloha_static_coffee/tree/b144896feb1f37398a862927b22cd3abdf005a6b (accessed 20 October 2026).